STANDARD OPERATING PROCEDURE

Power Tools Safety

Document No.: SOP-LV-004	**Revision:** 1.0	**Effective Date:** July 13, 2026	**Review Date:** July 13, 2027
Applies To: All field technicians and crews		**Jurisdiction:** Ontario / Manitoba	
Approved By: ____________________________		**Status:** Active	

1. Purpose

This Standard Operating Procedure (SOP) establishes the minimum safe work requirements for the selection, inspection, use, maintenance, and storage of portable power tools used by low-voltage and structured cabling technicians on commercial and institutional job sites, including schools. It is intended to reduce the risk of laceration, puncture, electrical shock, eye injury, and struck-by/caught-in incidents associated with corded power tools.

2. Scope

This SOP applies to all employees, subcontractors, and site personnel operating company-owned or company-supplied power tools, including but not limited to: drills, rotary hammer drills, impact drivers, reciprocating saws, chop/cut-off saws, and angle grinders. It applies to all job sites in Ontario and Manitoba, including occupied facilities such as schools where additional coordination with facility staff and building occupants is required.

3. References

- Ontario Occupational Health and Safety Act (OHSA) and O. Reg. 213/91 — Construction Projects
- CSA Standard Z462 — Workplace Electrical Safety
- CSA Standard Z460 — Control of Hazardous Energy (Lockout/Tagout)
- Canadian Electrical Code (CEC), Part I
- National Building Code (NBC) of Canada, applicable sections
- Manitoba Workplace Safety and Health Act and Regulation, where applicable
- WSIB reporting requirements (Ontario) / WCB reporting requirements (Manitoba)
- Manufacturer operating manuals and safety instructions for each specific tool
- SOP-LV-002 — Concrete Floor Drilling and Utility Strike Prevention
- SOP-LV-003 — Ontario Jobsite Health and Safety Checklist

4. Definitions

- Power Tool — Any hand-held or hand-operated tool powered by electricity (corded), pneumatic, or combustion source.
- GFCI — Ground Fault Circuit Interrupter; a device that disconnects power when it detects current leakage to ground.
- PPE — Personal Protective Equipment.
- Competent Worker — A worker who is qualified because of knowledge, training, and experience to perform the specific task safely, and who is familiar with applicable legislation and hazards.

- Guard — A fixed or adjustable physical barrier designed to prevent contact with a moving or cutting part of a tool.

5. Responsibilities

5.1 Supervisors / Site Leads

- Ensure only trained, competent workers are assigned power tools appropriate to the task.
- Verify tools brought on site are in good working condition prior to the start of work.
- Confirm GFCI protection is available and functional wherever corded tools are used.
- Coordinate with facility contacts (e.g., school administration) on scheduling, noise, and access restrictions before power tool use begins.
- Ensure incidents and near-misses involving power tools are reported per company incident reporting procedure.

5.2 Technicians

- Conduct a pre-use inspection of every power tool before each use, per Section 7.
- Use the correct tool and correct accessory (bit, blade, disc) for the task, per manufacturer specifications.
- Wear all required PPE for the task being performed.
- Report damaged, malfunctioning, or unsafe tools immediately and remove them from service.
- Complete and sign the acknowledgment section of this SOP prior to operating company power tools.

6. General Safety Requirements

1. Only competent, trained workers may operate power tools. New or unfamiliar tools require a manufacturer orientation before first use.
2. All corded electric tools used on site must be connected through a GFCI-protected outlet or portable GFCI device, in accordance with CEC and O. Reg. 213/91.
3. Tools must never be used with damaged cords, missing guards, cracked housings, or malfunctioning switches. Damaged tools are tagged out of service and removed from the work area immediately.
4. All factory-installed guards must remain in place and functional. Guards must not be removed, propped open, or bypassed.
5. Cords and hoses must be kept clear of walkways, ladders, and work areas, and must not create a trip hazard or contact sharp edges.
6. Tools must be unplugged from power before changing bits, blades, or accessories, and before clearing jams.
7. Required PPE — safety glasses, hearing protection where applicable, cut-resistant gloves for blade/bit handling, and steel-toe footwork — must be worn as appropriate to the tool and task.
8. Loose clothing, dangling jewelry, and unrestrained long hair must be secured before operating rotating or reciprocating tools.
9. Bystanders and other workers must be kept clear of the tool's operating radius; barricades or verbal warnings must be used in occupied or public areas such as school hallways.
10. Tools must be carried by the handle, never by the cord or hose, and must never be hoisted by the cord.

7. Pre-Use Inspection Checklist

A pre-use inspection is required before every shift and before switching between tools. Any item that fails inspection must be tagged and removed from service — do not use.

#	✓	Item
1	☐	Cord and plug free of damage, fraying, or exposed wire
2	☐	Housing free of cracks; no missing screws or panels
3	☐	Guard(s) present, undamaged, and move freely
4	☐	Trigger/switch operates smoothly and returns to off when released
5	☐	Bit, blade, or disc correctly rated for the tool and free of damage
6	☐	Bit/blade/disc securely tightened per manufacturer torque spec
7	☐	Chuck key or wrench removed before powering on
8	☐	GFCI protection confirmed and tested (corded tools)
9	☐	No unusual noise, vibration, heat, or smell on brief test run

8. Tool-Specific Requirements

8.1 Drills and Rotary Hammer Drills

- Confirm hole location has been cleared per SOP-LV-002 prior to drilling into concrete, block, or structural elements.
- Maintain a firm two-handed grip on rotary hammers and use the side handle where provided, to control torque reaction.
- Use dust extraction or wet methods where required for silica-generating tasks, per OHSA silica exposure requirements.

8.2 Saws (Chop Saws, Reciprocating Saws, Hole Saws)

- Confirm material is secured/clamped before cutting; never cut freehand material that can shift or bind the blade.
- Keep hands and body clear of the blade path; allow the blade to reach full speed before contacting material.
- Allow the tool to come to a complete stop before setting it down.

8.3 Angle Grinders

- Use only discs rated for the tool's rated speed (RPM); never exceed the disc manufacturer's maximum RPM.
- Face shield in addition to safety glasses is required when grinding or cutting metal.
- Position the body so the disc rotates away from you; be aware of kickback risk.

9. Occupied and Institutional Sites (e.g., Schools)

- Confirm power tool use windows with the school/facility contact in advance; avoid high-noise work during instructional periods where required by the site agreement.
- Barricade or cone off the immediate work area when operating tools in corridors or shared spaces.
- Be aware of concealed building services (in-floor heating, electrical, plumbing) before drilling or cutting into floors, walls, or ceilings — refer to SOP-LV-002 for utility strike prevention procedures.
- Report any accidental contact with a building service immediately to the site supervisor, regardless of apparent severity.

10. Maintenance and Storage

- Tools are wiped down, inspected, and returned to their case or designated storage location at the end of each shift.
- Damaged or malfunctioning tools are tagged "Out of Service" and reported to the supervisor for repair or replacement — they are not to be used or returned to circulation.
- Scheduled maintenance (e.g., brush replacement, blade/bit resharpening) is performed per manufacturer intervals.

11. Incident and Near-Miss Reporting

Any injury, near-miss, tool failure, or property/utility strike involving a power tool must be reported to the site supervisor immediately, regardless of severity. A preliminary incident report is to be completed the same day, consistent with company incident reporting procedure (see IR-2026-07-08-001 for reporting format). Damaged tools involved in an incident are removed from service and retained pending investigation.

12. Non-Compliance

Failure to follow this SOP may result in removal from the task, additional training requirements, or disciplinary action up to and including termination, consistent with company policy. Willful removal or defeat of a guard or safety device is treated as a serious safety violation.

13. Technician Sign-Off

I confirm that I have read, understood, and received training on this Power Tools Safety SOP (SOP-LV-004), and I agree to comply with all requirements described herein.

Technician Name (Print):	**Signature:**	**Date:**
Technician Name (Print):	**Signature:**	**Date:**
Supervisor Name (Print):	**Signature:**	**Date:**